

EC331: Analysis Plan

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1 Research question

I seek to investigate the long-run effects of trade liberalisation reforms on employment in India from 1991 onwards. I am particularly interested in if this effect was heterogeneous by gender, potentially offering insight into gender implications of sectoral reallocation of production and resources, and the relationship between changes in labour demand for men and women. Differential treatment effects by education levels and for rural/urban areas will also be considered.

My empirical approach relies on a Bartik instrument; in essence answering how the degree of exposure to trade liberalisation affects changes in employment, relying on the plausible exogeneity of trade reforms in India from 1991 - 1997, argued in line with existing literature (Topalova, 2005; 2010).

2 Data sources

- **Decennial census data:** 3 data points available; 1991, 2001 and 2011 (current focus is on the first difference between 1991 and 2001). Provides district-level employment rates with the number of workers per industry group, further split by rural/urban areas (one of the datasets used includes separate City/Urban Agglomeration observations, which are dropped for uniformity), gender and education levels. Similar data for marginal workers also available.
- My 1991 sample consists of 32 states and 451 districts. With the further splits by education group and rural/urban, there are 20 observations per district (9020 in total). I am currently in the process of cleaning and compiling 2001 data, for which there are 35 states and union territories, arising from 3 districts separating from states in 2000. Plan to keep to 1991 categorisation of districts.
- **Trade data by industry:** World Integrated Trade Solution (WITS) is a trade software provided by the World Bank to access international trade databases. Can see tariffs applied by product (some concern over sensitivity of assigning 'ambiguous' products to industry groups).

3 Empirical specification and identification

My main specification, taking first differences, is as follows:

$$\Delta y_{dt} = \alpha + \beta \Delta \text{Tariff}_{dt} + \gamma_t + \Delta \varepsilon_{dt}$$

where

$$\Delta \text{Tariff}_{dt} = \sum_j w_{1991,dj} \cdot \Delta \text{Tariff}_{j,1991-1997}$$

instrumenting for the change in tariff exposure for each district from 2001 to 1991. My Bartik instrument approach assumes exogeneity of the shifts - tariff changes are as-good-as-randomly assigned to industries, and they only affect district employment through the treatment. Instrument at present excludes tariff changes from 1997 to 2001, as evidence from Topalova suggests tariffs in this period were correlated with industry characteristics. This would potentially violate exogeneity if tariff changes systematically differ in industries concentrated in certain regions with for example, labour supply shocks. w represents the employment share (predetermined before trade reforms to navigate simultaneity) for industry j in district d . y_{dt} , my outcomes of interest, correspond to employment rates and labour force participation (overall, and specific demographic groups).

In line with Borusyak et al (2024), I aim to estimate standard errors through an 'exposure-robust' approach. This accounts for districts with similar industry shares (and so similar instrument values) may have correlated error terms due to common exposure to unobserved shocks.

Further specifications include adding an interaction term with a dummy for if labour market regulation in the district is classified as pro-worker, as per Besley and Burgess (2004) to see if this influenced how outcomes were differentially affected by trade reforms.

4 Hypotheses

- **Employment outcomes overall;** $H_0: \beta = 0$ - changes in exposure to tariffs have no statistically significant effect on decadal changes in district employment outcomes. $H_A: \beta \neq 0$ - statistically significant effect.
- **Employment outcomes by different groups;** Motivated by literature - expect less educated workers to benefit least/experience reduced employment from trade liberalisation. For women's employment - could be argued both ways, potentially converse relationship to male employment patterns through many women's labour force participation decisions in developing countries seeking to 'smooth' household employment.

5 Robustness checks

- An important consideration is changes to other trade protection measures, such as delicensing, regulation of FDI flows. Although suggested in the literature that tariff changes in India are correlated with these, will try and access data on alternative trade protection measures.
- Robustness checks for identifying assumption: ideally, would regress lagged employment outcomes on instrument, however difficulty in accessing employment data before 1991 (might try NSS). Balance tests for the instrument, using proxies for potential confounders.

6 Current concerns and areas to focus on

- Finding data on agricultural output by crop for districts (potentially by proxy with soil and rainfall data) to account for non-tradeable crops to better capture treatment variation by districts.
- Statistical power; having sufficient variation in the instrument - pre-det industry shares appear to have sufficient heterogeneity, but will compute final instrument values and consider.
- Building motivation and hypotheses with trade theory/models in the absence of perfect mobility of factors across geographical regions. Household models of labour market decisions in developing countries.

References

Besley, T. and Burgess, R. (2004) ‘*Can labor regulation hinder economic performance? Evidence from India*’, The Quarterly Journal of Economics, 119(1), pp. 91–134. doi:10.1162/003355304772839533.

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Topalova, P. (2010) ‘*Factor immobility and regional impacts of trade liberalization: Evidence on poverty from India*’, American Economic Journal: Applied Economics, 2(4), pp. 1–41. doi:10.1257/app.2.4.1.